Gated and Axis-Concentrated Localization Network for Remote Sensing Object Detection

Xiaoqiang Lu, *Senior Member, IEEE*, Yuanlin Zhang, Yuan Yuan, *Senior Member, IEEE*, and Yachuang Feng

***Abstract*— In the multicategory object detection task of high-resolution remote sensing images, small objects are always difficult to detect. This happens because the influence of location deviation on small object detection is greater than on large object detection. The reason is that, with the same intersection decrease between a predicted box and a true box, Intersection over Union (IoU) of small objects drops more than those of large objects. In order to address this challenge, we propose a new localization model to improve the location accuracy of small objects. This model is composed of two parts. First, a global feature gating process is proposed to implement a channel attention mechanism on local feature learning. This process takes full advantages of global features' abundant semantics and local features' spatial details. In this case, more effective information is selected for small object detection. Second, an axis-concentrated prediction (ACP) process is adopted to project convolutional feature maps into different spatial directions, so as to avoid interference between coordinate axes and improve the location accuracy. Then, coordinate prediction is implemented with a regression layer using the learned object representation. In our experiments, we explore the relationship between the detection accuracy and the object scale, and the results show that the performance improvements of small objects are distinct using our method. Compared with the classical deep learning detection models, the proposed gated axis-concentrated localization network (GACL Net) has the characteristic of focusing on small objects.**

***Index Terms*— Deep learning, gated axis-concentrated localization network (GACL Net), localization, remote sensing, small object detection.**

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I. Introduction

REMOTE sensing object detection is a hot-topic problem that has been investigated for decades [1]. The task of remote sensing object detection is to detect specific categories of objects in the images. This task has been applied in road detection [1], airport detection [2], car detection [3], and so on.

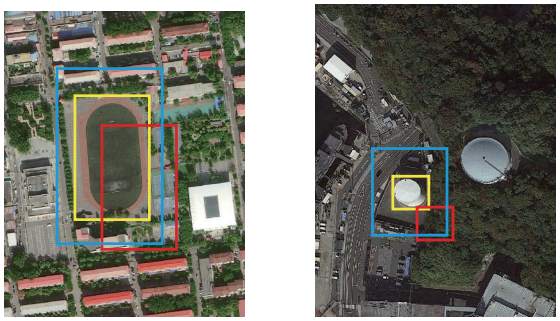
For candidate boxes containing potential objects, localization can greatly affect the detection results for small objects. Recently, the judgment of the correct position is based on the Intersection over Union (IoU) between the real bounding box and the predicted bounding box. After several pixels' deviations, the intersection between the prediction and the ground truth decreases fast to zero for small objects, so does IoU. As shown in Fig. 1, the yellow bounding box is the ground truth, the blue one indicates width or height deviation, and the red one represents the center coordinate deviation. These deviation values are the same between Fig. 1(a) and (b). It is obvious that the red box of the small object is totally an error result, while the red box of the large object is tolerable. For the blue box, there is a similar phenomenon. This is because the ratio of intersection area over union area drops faster as the union is smaller.

Therefore, it is necessary to propose a detector for improving the localization of small objects. For the localization task, most of the recent remote sensing detectors [2], [4], [5] adopt global features, which focus on semantic information. However, global features contain little spatial detail. Hence, it is necessary to address this problem by making the detector consider the spatial details of local features during localization. In this article, a novel regression model is proposed to take the global feature as semantic guiding for the local feature.

The proposed localization regression model is embedded in a convolutional neural network (CNN) object detection framework. The most significant part of this regression model is embedding the traditional global features into the local features in the form of gating. Specifically, fully connected layers of the CNN framework represent global information, and the pixels of the convolutional feature cube stand for

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(a) (b)

Fig. 1. Illustration of the worse effect of location deviation on small object detection. The yellow bounding boxes are the ground truth boxes. The blue ones have the same centers but different widths (or heights) compared with the ground truths. The red ones have the same widths (or heights) but different centers compared with the ground truths. (a) Large object. (b) Small object. It can be seen that with the same deviations (in center positions or in widths/heights), bad boxes are more intolerable for small objects.

local features. Moreover, it should be emphasized that fully connected layers still preserve part of spatial information, for what the traditional regression model can get the bounding box deviation from the global feature [6]. In comparison, the local feature provides more detailed spatial information [7]. In addition, as mentioned by Hu *et al.* [8], feature response recalibration between channels by using global features is favorable for object detection systems. Therefore, it is reasonable to focus on both the global features and the local features for better localization, which can take their complementary advantages. This article proposes a global feature gate (GFG), which implements a channelwise gating through multiplying the global feature by the local feature. Another crucial component of this regression model is the axis-concentrated prediction (ACP), which is designed to avoid disturbance from the spatially orthogonal direction in localization. Especially, the ACP is implemented by concentrating a 3-D feature cube to two 2-D horizontal and vertical feature planes, where the two dimensions, respectively, indicate the horizontal/vertical dimension and the channel dimension. Subsequently, to compute coordinate deviations, the horizontal feature plane is used for predicting the deviation of object center coordinate x and object width w, and the vertical feature plane is used for predicting the deviation of object center coordinate y and object height h.

In order to implement the proposed localization regression model, a CNN named gated axis-concentrated localization network (GACL Net) is proposed as shown in Fig. 2. This localization network is trained together with a feature extractor and a recognizer, making the detection pipeline end-to-end and easier to use.

To evaluate the proposed model, experiments are conducted on different multicategory remote sensing object detection data sets with different CNN architectures (i.e., CaffeNet [9], Visual Geometry Group (VGG) [10], or VGG variation). The results show that our model can significantly improve the average precision (AP) on small objects.

In summary, the main contributions of this article are as follows.

1) In this article, a CNN-based network named GACL Net is proposed to improve the localization accuracy of object categories with small scales. Thus, the GACL Net can address the small object detection and provide more accurate localization.
2) The GFG is proposed to take full advantages of global features' abundant semantics and local features' spatial details. The GFG takes the global feature as a semantic guiding for the local feature. It makes up for local feature shortage in small object localization by avoiding a noncorrespondence between the box and the recognition result.
3) The proposed ACP, which extremely compresses the feature in the direction unrelated with the current coordinate, can avoid coordinate prediction disturbance and get more accurate localization.

The remainder of this article is organized as follows. In Section II, the related works of remote sensing object detection are reviewed. In Section III, our object detection framework is described. In Section IV, the proposed localization model is introduced. In Sections V and VI, the experimental results and the conclusions are discussed, respectively.

II. Related Works

As an important part of the remote sensing image analysis, object detection has been paid much attention. This section reviews the previous object detection frameworks, which have gone through three stages of development: low-level feature stage, midlevel feature stage, and deep learning stage [11]. In addition, we have introduced several existing methods about improving localization accuracy in both the remote sensing image analysis and the natural image analysis.

A. Object Detection Frameworks

1) Low-Level Feature Stage: As aforementioned, the first stage of the object detection framework is the local feature stage. Lowe [12] proposed a method named scale-invariant feature transform (SIFT), which extracts the interested points of objects and then matches these points with those in other images. The SIFT feature is insensitive to scaling, panning, and rotation. Li *et al.* [13] introduced the SIFT to remote sensing object detection. Han *et al.* [14] made further research of the SIFT to cooperate with deep learning methods. Dalal and Triggs [15] proposed a histogram of oriented gradients (HOG), where the object is represented by the gradient intensity and distribution of gradient direction. Grabner *et al.* [3] introduced the HOG to the remote sensing analysis. Cheng *et al.* [16] proposed a multilevel HOG feature pyramid called the collection of part detectors (COPD). Zhang *et al.* [17] utilized the saliency analysis for remote sensing images. Han *et al.* [18] proposed an attention model, which is named as the Fisher discrimination dictionary learning (FDDL), to enhance the saliency of objects.

2) Midlevel Feature Stage: The second stage is the midlevel feature stage. Zhang *et al.* [19] proposed a rotation-invariant part-based model for remote sensing object detection. Li *et al.* [20] proposed a contour-based spatial model.

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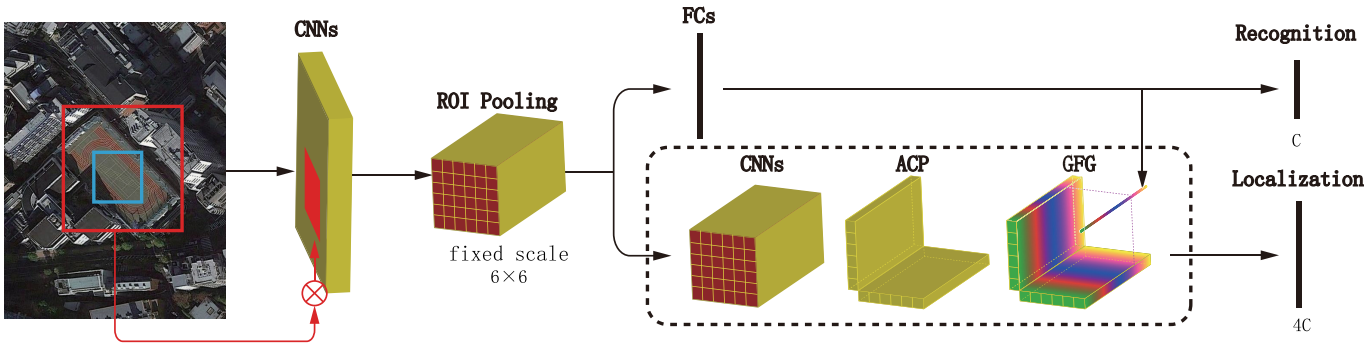


Fig. 2. Visualization of the GACL Net. In Section III, we propose ACP and GFG structures to improve small-scale object detection. In this figure, C means the count of object categories, and the length of final feature $4C$ includes four coordinates for every category.

Fei-Fei and Perona [21] proposed a bag-of-words (BoW) model, another kind of midlevel object detection feature. This model is simple but efficient and is invariant to viewpoint changes and background clutter [22]. Sun *et al.* [23] proposed a spatial-sparse-coding BoW (SSCBoW).

3) Deep Learning Stage: The current stage is of deep learning, which was started in 2006, when Hinton and Salakhutdinov [24] made a breakthrough in the deep learning field. According to whether there is a region proposal mechanism or not, the deep learning object detection methods in nature scene were divided into *two-stage detections* and *one-stage detections* [25], respectively.

a) Two-stage detections: The two-stage detection pipeline consists of category-agnostic proposals and category-specific detections. Sermanet *et al.* [26] proposed an *OverFeat* that applies the CNN to integrated recognition, localization, and detection. Most importantly, Girshick *et al.* [6] first proposed an ingenious deep learning detection framework, namely region-based CNN (R-CNN). He *et al.* [27] proposed a spatial pyramid pooling network (SPP-Net) to eliminate the requirement of a fixed-size (e.g., 224 × 224) input image of previous CNNs. Girshick [28] then proposed a Fast R-CNN that takes the advantage of SPP-Net and OverFeat. Ren *et al.* [25] proposed a Faster R-CNN. In this article, a region proposal network (RPN) was introduced to integrate the region proposal process into the CNN model. Wang *et al.* [29] proposed the A-Fast-RCNN, which learns an adversarial network that generates examples with occlusions and deformations.

b) One-stage detections: The one-stage detection pipeline only contains class-specific detection. Redmon *et al.* [30] first proposed a one-stage detection method named as you only look once (YOLO). Then, Liu *et al.* [31] proposed another one named as *single-shot multibox detector* (SSD), which considers multiscale regional features. Furthermore, Fu *et al.* [32] proposed a deconvolutional single-shot detector (DSSD), which takes a skip connection with the deconvolutional layers to improve representational capacity. Lin *et al.* [33] proposed Retina-Net for dealing with the problem of easy example dominancy.

In the remote sensing field, most of the current deep learning object detection methods are two-stage detections, since they serve better accuracy [25] compared with the one-stage detections. Zhu *et al.* [34] proposed a CNN-based orientation robust feature extractor. Cheng *et al.* [4] proposed a rotation-invariant CNN (RICNN) framework with a rotation-invariant fully connected layer. In addition, Hu *et al.* [35] considered the convolutional feature coding of multiscale images to address the scaling problem.

B. Localization Models

As illustrated in the YOLO [30], the localization error is one of the major errors in current object detection methods, whether it may be two-stage detection or one-stage detection. Several works to address the localization problem are listed in the following.

In the nature scene detection field, Gidaris and Komodakis [36] proposed a framework named as multiregion deep CNN (MR-CNN), which detects objects with high localization accuracy through different location-sensitive appearance factors of the same object. Gidaris and Komodakis [37] proposed a localization model (LocNet) that replaces the previous localization regression model with a probability model of horizontal and vertical coordinates.

In the remote sensing detection field, Long *et al.* [11] proposed an accurate localization model for detected objects in the field of remote sensing image processing. This article follows previous deep learning frameworks to generate bounding box regression results and applies the nonmaximum suppression (NMS) method to reduce the redundancy of overlapped bounding boxes. Then, an unsupervised score-based bounding box regression (USB-BBR) algorithm was proposed to further reduce these redundant boxes after refining. Obviously, this article pays more attention on the postprocessing of detection results rather than on the localization regression model itself.

However, these methods have problems in detecting small objects. Thus, we propose a novel CNN framework to address this problem by presenting an accurate localization model.

III. Object Detection Framework

This article uses the classic CNN detection pipeline, which consists of several parts: region proposal generation, CNN feature extraction, region of interest (ROI) pooling, category recognition, and location regression (i.e., localization).

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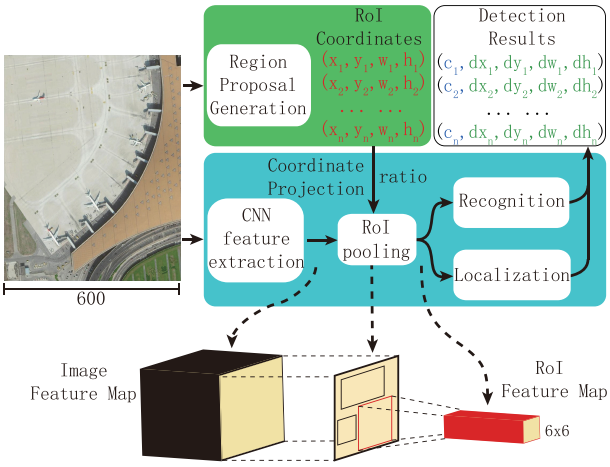


Fig. 3. Overall framework of the detection system adopted in this article. Green block: region proposal method that generates ROI coordinates for the blue block. Blue block: end-to-end detection model that consists of CNN, ROI pooling, recognition, and localization.

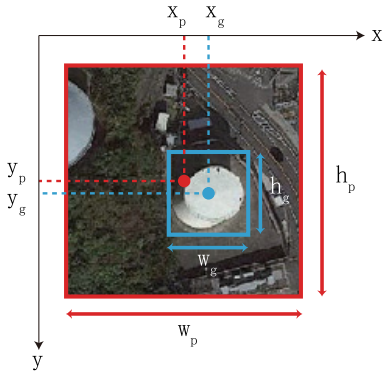


Fig. 4. Illustration of the concept of coordinate refinements. The blue box is the ground truth and the red one is the coarse proposal. To predict more accurate box, the refinement between the proposal and the ground truth is predicted.

The overall framework is shown in Fig. 3. In this figure, the green block represents a region proposal method that generates ROI coordinates. The blue block is the main branch of an object detection framework, which predicts object categories and the refinements of the generated ROI coordinates. Starting from CNN feature extraction, this network gets ROI feature maps by implementing ROI pooling and obtains detection results by applying recognition and localization. Details of these parts in Fig. 3 are listed as in the following.

1) *Region Proposal Generation:* The selective search [38] (SS) method is used to predict the candidate bounding boxes (namely ROIs). This method considers the similarity of hue, texture, scale, and overlap. In our system, the SS method is used to generate a set of ROI bounding box coordinates $\mathbf{CORD} = \{\boldsymbol{cord_i}\}_{i=1}^{N}$ for later processing. And $\boldsymbol{cord_i} = (x_i, y_i, w_i, h_i)$.
2) *CNN Feature Extraction:* The CNN input is the cropped image during the training phase and the entire image during the test phase. The entire image is resized to keep its short axis with a fixed size. In addition, in each of the convolutional layers, given an input image or a feature map $\mathbf{I} = \{\boldsymbol{I^s}\}_{s=1}^{S}$ with S channels, the convolutional layer computes an output $\mathbf{O} = \{\boldsymbol{O_c}\}_{c=1}^{C}$ with C channels by

$$\boldsymbol{O_c} = \mathbf{K_c}(\boldsymbol{I}) = \sum_{s=1}^{S} \boldsymbol{\kappa_c^s} * \boldsymbol{I^s} + \boldsymbol{b_c} \tag{1}$$

where $\mathbf{K_c}$ represents the convolutional kernel $[\boldsymbol{\kappa_c^1}, \boldsymbol{\kappa_c^2}, \ldots, \boldsymbol{\kappa_c^s}, \ldots, \boldsymbol{\kappa_c^S}]$ for output channel c and $*$ is a convolutional operation. The values of $\boldsymbol{\kappa_c^s}$ are weights and the values of $\boldsymbol{b_c}$ are biases in the convolutional kernel.
3) *ROI Pooling:* First, given some SS-produced ROI bounding box coordinates $\mathbf{CORD} = \{\boldsymbol{cord_i}\}_{i=1}^{N}$, we get rescaled coordinates $\mathbf{CORD}' = \{\boldsymbol{cord_i}'\}_{i=1}^{N}$ through a projection from the image scale to the CNN feature map scale

$$\begin{aligned} \boldsymbol{cord_i}' &= \boldsymbol{cord_i} \cdot Sc \\ &= (x_i \cdot Sc, y_i \cdot Sc, w_i \cdot Sc, h_i \cdot Sc) \end{aligned} \tag{2}$$

where Sc denotes the ratio of spatial scaling. Then, as shown in Fig. 3, ROI feature maps $\mathbf{R}' = \{\boldsymbol{R^{i\prime}}\}_{i=1}^{N}$ are clipped from the feature map $\mathbf{O} = \{\boldsymbol{O_c}\}_{c=1}^{C}$ (only in the test phase) using $\mathbf{CORD}'$ and then resized to a fixed scale to get $\mathbf{R} = \{\boldsymbol{R^i}\}_{i=1}^{N}$.
4) *Recognition:* This paragraph evaluates different category confidence probabilities for input ROIs. Taking the ROI feature maps $\mathbf{R} = \{\boldsymbol{R^i}\}_{i=1}^{N}$ as the input, the recognition model utilizes the softmax activation layer after several fully connected layers to generate category probabilities

$$\mathbf{P} = \{\mathbf{p}^i\}_{i=1}^{N} = \left\{p_c^i\right\}_{i=1,c=1}^{N,C}, \quad p_c^i \in [0, 1] \tag{3}$$

where i and c denotes the ROI index and the potential category index, respectively.
5) *Localization:* This part predicts different category coordinate refinements for an input ROI. The illustration of the concept of refinements is shown in Fig. 4. The blue box is the ground truth and the red one is the coarse proposal. To predict more accurate boxes, a refinement between the proposal and the ground truth is predicted. With the ground truth bounding box (x_g, y_g, w_g, h_g) and the coarse proposal bounding box (x_p, y_p, w_p, h_p)

$$dx = (x_g - x_p)/w_p \tag{4}$$
$$dy = (y_g - y_p)/h_p \tag{5}$$
$$dw = \log(w_g/w_p) \tag{6}$$
$$dh = \log(h_g/h_p). \tag{7}$$

During the test phase, the network outputs the bounding box refinements (dx, dy, dw, dy), which are used to compute the predicted box $(\tilde{x}_g, \tilde{y}_g, \tilde{w}_g, \tilde{h}_g)$ according to the antiformula of (4)–(7). Taking the ROI feature maps $\mathbf{R} = \{\boldsymbol{R^i}\}_{i=1}^{N}$ as the input, the proposed localization

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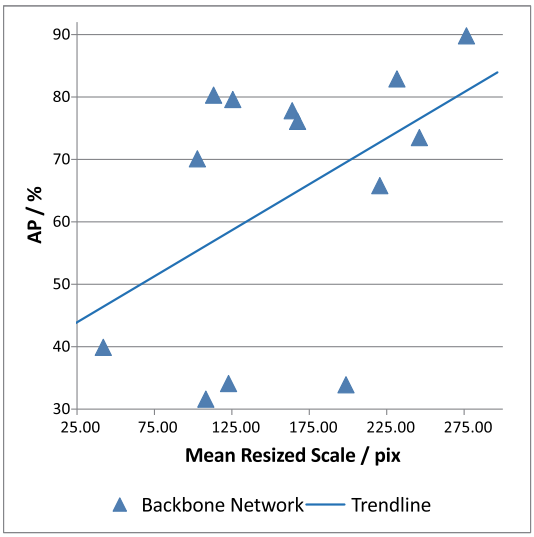


Fig. 5. Relationship between AP and mean resized scale. The AP is got by implementing Fast R-CNN on the HRRSD data set. The mean resized scale, in pixels, is the square root of the object area after adjusting the width and the height strictly according to the network structure.

model generates refinements $\mathbf{B}$

$$\mathbf{B} = \{\mathbf{b}^i\}_{i=1}^{N} = \left\{\mathbf{b}_c^i\right\}_{i=1,c=1}^{N,C} \tag{8}$$

$$\mathbf{b}_c^i = \left(dx_c^i, dy_c^i, dw_c^i, dh_c^i\right) \tag{9}$$

for different categories $c = 1, 2, 3, \ldots, C$.

In addition, for a candidate box i, its refinements $\mathbf{b}^i = \{\mathbf{b}_c^i\}_{c=1}^{C}$ are filtered by the best scored category $c_i = \arg\max_c(p_c^i)$ to reserve only one of them $\mathbf{b}^{i'} = \mathbf{b}_c^i|(c = c_i)$. Thus, the detection results are $\mathbf{D} = \{\mathbf{d}_i\}_{i=1}^{N}$ where

$$\mathbf{d}_i = (c_i, \mathbf{b}^{i'}) = (c_i, dx_i, dy_i, dw_i, dh_i). \tag{10}$$

IV. Proposed Localization Model

The GACL-Net is a localization model to address the problem of small object localization. We illustrate the relationship between the detection AP of an object category and its mean resized scale in Fig. 5. This AP is got by implementing the Fast R-CNN [28] on a remote sensing data set high-resolution remote sensing detection (HRRSD) [39][1] with 13 categories for object detection, which is built by ourselves and is category-balanced. The mean resized scale, in pixels, is the square root of the area of a resized object bounding box and will be explained in detail in Section V-C. According to the trend line, it can be indicated that the smaller an object is, the more difficult it is to detect.

To address the aforementioned problem, we propose to utilize global information to guide the channel attention mechanism of the local information. On the one hand, the local information can tell us in detail where the object is [40]. On the other hand, the global information from layers can adjust channelwise local feature responses to emphasize or suppress a channel according to whether it is informative or irrelevant, respectively [8].

[1]https://github.com/CrazyStoneonRoad/TGRS-HRRSD-Dataset

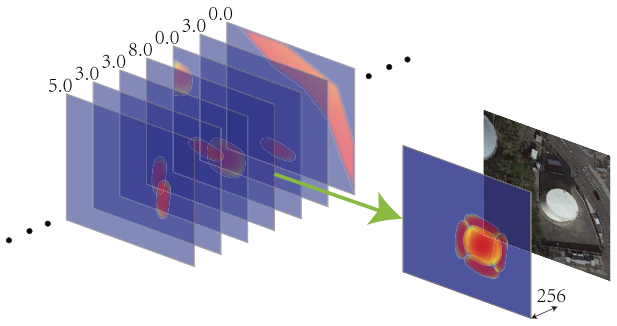


Fig. 6. Architecture of global feature gating. All pixels of a feature map channel are scaled by the same number.

A. Global Feature Gating

It is exactly the aforementioned channel attention mechanism of the local information which is designed to better detect small objects. In this section, this channel attention mechanism is distinctly presented and discussed.

1) Presentation: As shown in Fig. 6, during the global-gating processing, pixels in the same feature map channel share a common multiplier, which comes from the global feature vector. The global feature from fully connected layer can grasp semantic information of the whole ROI. This global feature can adaptively choose informative channels and remove unnecessary ones [8].

With the fully connected feature f_{cn}

$$f_{cn} = \{f_{cn}(l)\}_{l=1}^{L} \tag{11}$$

as global information and the convolutional feature $\boldsymbol{F}_{cv}$

$$\boldsymbol{F}_{cv} = \{F_{cv}(i, j, l)\}_{i=1,j=1,l=1}^{N,N,L} \tag{12}$$

as local feature, the global feature gating results $\boldsymbol{F}_{gfg}$ are computed by

$$\boldsymbol{F}_{gfg} = \{F_{gfg}(i, j, l)\}_{i=1,j=1,l=1}^{N,N,L} \tag{13}$$

$$F_{gfg}(i, j, l) = f_{cn}(l) \cdot F_{cv}(i, j, l). \tag{14}$$

It should be noted that the length of f_{cn} is the same as the channel number of $\boldsymbol{F}_{cv}$, thus a dimensionality reduction is required before GFG.

2) Discussion: The global feature from fully connected layer can grasp better attribute information of the entire ROI which indicates the common attributes of all the local areas. And these common attributes can be used to guide the selection of local area attributes so that the model can retain useful information and filter out useless information.

In this article, the local area is a pixel of a convolutional feature map. In general, for the convolutional feature, different channels can be taken as different variation factors or attributes [41]. Thus, the common variation factors of local features hidden in the fully connected layer can guide the convolutional feature.

B. Axis-Concentrated Prediction

This section describes the proposed localization model, named the ACP. It is designed to avoid coordinate prediction

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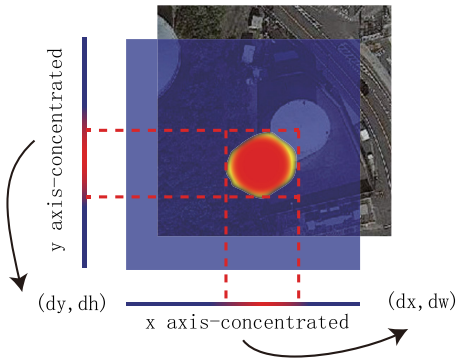


Fig. 7. Architecture of ACP which can avoid object coordinate prediction disturbance from the unrelated direction. For simplicity, only one of the feature map channels is shown. This ROI feature map is first concentrated on the two natural axes. Then, the axis-concentrated features are used to predict bounding box coordinate refinements in the corresponding direction.

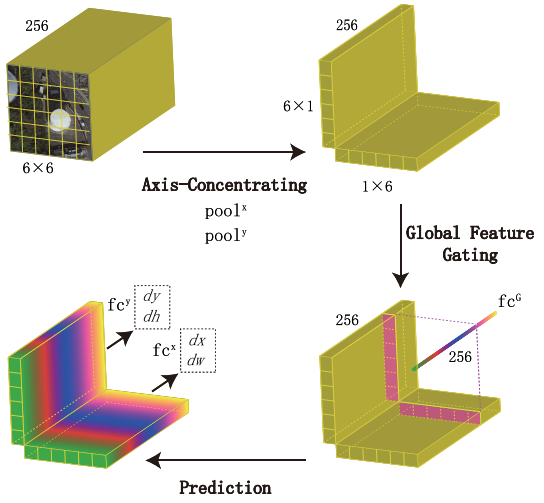


Fig. 8. Localization model with reasonably arranged order of the global feature gating and the ACP. Axis concentrating is launched first to avoid coordinate prediction disturbance and gating is then utilized to select appropriate channels before prediction.

disturbance from the unrelated direction. When calculating horizontal coordinates, local features with different vertical coordinates are fed in, thereby resulting in coordinate prediction disturbance. Therefore, the ACP is proposed to address this problem.

1) Presentation: As shown in Fig. 7, in order to avoid coordinate prediction disturbance from the unrelated direction, the information of the ROI feature map is first concentrated on two natural axes, i.e., the horizontal and the vertical. Then, the axis-concentrated features are used to predict bounding box coordinate refinements in their corresponding directions. For example, the x-axis-concentrated feature, where vertical pixels are compressed, is used to predict (dx, dw).

To implement the concentrating operation, max pooling is adopted and can be replaced by other pooling operations. Taking the nth channel as $\mathbf{F^n}$ and the feature map coordinates as (x, y), the concentrating operation can be formulized as follows:

$$\mathrm{Conc}_x(\mathbf{F^n}) = \max_y(F^n(x, y)) \tag{15}$$

$$\mathrm{Conc}_y(\mathbf{F^n}) = \max_x(F^n(x, y)). \tag{16}$$

2) Discussion: This operation can largely cut the computational cost. To illustrate that, we take the feature map size as (N, N, C) and the number of categories as Cls. Considering the coordinate refinement prediction, the original computational cost is

$$(N \times N \times C) \times (Cls \times 4) = 4 \times Cls \times N^2 \times C \tag{17}$$

and the axis-concentrated computational cost is

$$(1 \times N \times C) \times (Cls \times 2) + (N \times 1 \times C) \times (Cls \times 2) = 4 \times Cls \times N \times C. \tag{18}$$

It can be seen that the axis-concentrated operation can save $(N-1)/N$ of the original computational cost. Moreover, this operation would also affect the computational cost of the gating process.

C. GACL Net Architecture

As the GFG and the ACP have been illustrated, an organic combination of them is designed. To be specific, the ACP is launched first to avoid coordinate prediction disturbance from the unrelated direction and the GFG is utilized then to select appropriate channels before prediction.

1) Localization Model: As shown in Fig. 8, the GFG architecture is implemented between the axis-concentrating process and the coordinate refinement predicting process.

The proposed localization model contains three parts. First, given a convolutional feature map within an ROI, we concentrate it on the horizontal axis and the vertical axis to avoid coordinate prediction disturbance. Second, the GFG architecture is embedded in both the horizontal axis and the vertical axis. For these two axes, the same global feature vector is adopted to implement the GFG architecture. This setting would be verified later in Section V-G. Finally, the global-gated and axis-concentrated localization (GACL) feature planes are fed in regressors to predict coordinate deviations. The x-axis plane is fed for predicting (dx, dw) and the y-axis plane is fed for predicting (dy, dh).

2) Network Architecture: To carry out the proposed GACL Net, we take the Fast R-CNN as the backbone network. The whole architecture is visualized in Fig. 2.

Given an image $\mathbf{I}$, a candidate box, and a corresponding true box, this architecture starts by extracting the convolutional features from $\mathbf{I}$ by CNNs. To reduce the experimental difficulty, the CaffeNet [9] CNN network is preferred. The CaffeNet consists of five convolutional layers.

The CNN is then followed by an ROI pooling layer which has been introduced in Section III. This process involves scaling the input proposal box at a specific ratio and cropping in the "conv5" feature map using the rescaled proposal box. As a result, we get the ROI feature map.

The recognition part is one of the two branches after ROI pooling and contains three fully connected layers. The final

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layer outputs a vector with the length of C, i.e., the category number (including the background category).

The localization part is the other branch. First, the input ROI feature map is processed by two convolutional layers to access larger receptive field. Second, the ACP is implemented through two pooling layers. Third, with a vector obtained from the fully connected layer $\mathbf{fc^G}$, the gating process GFG is implemented through channelwise multiplication. Finally, two fully connected layers $\mathbf{fc^x}$ and $\mathbf{fc^y}$ are utilized to get coordinate refinements in corresponding direction, and their results are concatenated to form a vector of length $4C$.

3) Training: During the training process, the network tries to learn the localization model and the recognition model that the map ROI feature $\boldsymbol{R^i}$ to box coordinate refinements $\boldsymbol{B^i}$ and category possibilities $\boldsymbol{p^i}$, respectively. This article adopts the multitask loss function in [28]

$$L(\mathbf{p}^i, u, \mathbf{b}^i, \hat{\mathbf{b}}^i) = L_{\text{rec}}(\mathbf{p}^i, u) + \lambda[u \geq 1]L_{\text{loc}}(\mathbf{b}^i, \hat{\mathbf{b}}^i) \quad (19)$$

where the first part and the second part are the recognition and the localization loss function, respectively. For the ith proposal, $\mathbf{p}^i$ and $\mathbf{b}^i$ represent the predicted category probability vector in (3) and the predicted localization vector in (8), respectively. Moreover, u is the ground truth category of the proposal and $\hat{\mathbf{b}}^i$ represents the ground truth localization vector. If the proposal actually is background, i.e., $u = 0$, λ is set to 0. The recognition and the localization loss functions are formulated as follows:

$$L_{\text{rec}}(\mathbf{p}^i, u) = -\log \mathbf{p}^i_u = -\log \frac{e^{p^i_u}}{\sum_{c=1}^{N} e^{p^i_c}} \quad (20)$$

$$L_{\text{loc}}(\mathbf{b}^i, \hat{\mathbf{b}}^i) = \sum_{t \in \{x,y,w,h\}} \text{SmoothL1}\left(t^i_u - \hat{t}^i_u\right) \quad (21)$$

where SmoothL1 indicates a loss function designed for localization [28] and is formulated as follows:

$$\text{SmoothL1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \quad (22)$$

V. Experimental Results

A. Data Sets

To evaluate our localization model, experiments on two multicategory data sets are carried out. One of them is the HRRSD [39] with 21 761 color images acquired from Google Earth satellite imageries, which have the ground resolution from 0.15 to 1.2 m. The annotations of this data set contain 13 categories, while each of them has 3700–5000 objects. The object numbers and *object scales* of different categories are listed in Table I. The conception of object scale is illustrated later in Section V-C2. The "mean resized scale" column and the "resized scale std" column, respectively, indicate the average and the standard deviation of all the objects in a specific category.

Another data set is the NWPU VHR-10, which contains 800 color images. In these images, 715 are acquired from Google Earth satellite imageries with the resolution from 0.5 to 2 m. Other 85 images are acquired from Vaihingen data [42] with spatial resolution of 0.08 m. The manual annotations of this data set involves ten categories. The object numbers of all the categories are listed in Table II.

TABLE I
DIFFERENT CATEGORY OBJECT NUMBERS IN HRRSD

	number	Mean Resized Scale /pixel	Resized Scale Std /pixel
airplane	4899	113.21	67.98
baseball diamond	4033	231.61	117.85
basketball court	3876	108.19	57.46
bridge	4570	246.10	110.53
crossroad	4023	220.54	59.24
ground track field	3734	276.50	100.65
harbor	3884	163.96	94.16
parking lot	4958	122.85	54.45
ship	3886	167.44	110.37
storage tank	4406	125.60	68.41
T junction	4420	198.71	54.88
tennis court	4295	102.71	38.80
vehicle	4756	41.96	9.99

TABLE II
DIFFERENT CATEGORY OBJECT NUMBERS IN NWPU VHR-10

	number	Mean Resized Scale /pixel	Resized Scale Std /pixel
airplane	757	67.67	19.39
baseball diamond	390	84.26	22.95
basketball court	159	93.80	33.79
bridge	124	152.74	45.80
ground track filed	163	275.19	51.81
harbor	224	99.35	23.54
ship	302	52.45	14.83
storage tank	655	39.57	9.47
tennis court	524	62.60	15.50
vehicle	477	41.74	9.61

TABLE III
STRUCTURAL DETAILS OF OUR FRAMEWORK

		size
CaffeNet	conv1	$11 \times 11 \times 96, s4$
	pool1	$3 \times 3, s2$
	conv2	$5 \times 5 \times 256, s1$
	pool2	$3 \times 3, s2$
	conv3	$3 \times 3 \times 384, s1$
	conv4	$3 \times 3 \times 384, s1$
	conv5	$3 \times 3 \times 256, s1$
ROI Pooling		6×6
Recognition	fc6	4096
	fc7	4096
	fc8	14
Localization	conv6	$3 \times 3 \times 256, s1$
	conv7	$3 \times 3 \times 256, s1$
	fc^G	256
	fc^x	28
	fc^y	28

B. Experimental Setup

The detailed parameters of our network are shown in Table III. In this table, the kernel sizes of convolutional layers and the output shapes of fully connected layers are shown. The ROI pooling layer outputs shape-fixed feature maps. In addition, the ACP outputs a $1 \times 6 \times 256$ feature and a $6 \times 1 \times 256$ feature.

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TABLE IV
TRAINING PARAMETERS

parameter	value
base lr	0.001
γ	0.1
step size	60k
momentum	0.9
weight decay	0.0005
vertical flip	false
iterations	160k
mini batch	128

Some training hyperparameters are shown in Table IV. In this table, "base lr" is the base learning rate which decays by $(1-\gamma)$ every "step size." The "momentum" parameter is the weight of the previous update gradient. The "weight decay" is the regular term weight to constrain the training model size and to prevent overfitting. The training process will repeat "iteration" steps. In each step, "mini batch" ROIs are fed in the model.

During the test period, an NMS method is used to reduce the redundant boxes. The NMS first selects the box with the highest score among all the boxes. Then, taking τ as the NMS threshold, compared with the selected box, other boxes with IoU $>= \tau$ are removed. Finally, these two steps are repeated among those unselected boxes until all the input boxes are selected or removed. The NMS threshold τ is 0.3 in our experiments.

C. Evaluation Matrix

To clearly exhibit the relationship between the object size and the performance improvement of our method, we design an *improvement–scale diagram* according to [43]. The *improvement* is for the AP and the *scale* denotes the *object scale*. The definitions of the AP and the *object scale* are shown in the following.

1) Average Precision: To introduce the AP, some priori knowledge is required. First and foremost, we use TP, FP, and FN to represent the numbers of true-positive samples, false-positive samples, and false-negative samples, respectively.

Then, *Precision* and *Recall* are computed by

$$\text{Precision} = \frac{TP}{TP+FP} \tag{23}$$

$$\text{Recall} = \frac{TP}{TP+FN}. \tag{24}$$

At this point, the *Precision–Recall curve* (PRC) can be easily understood, since *Precision* and *Recall* are exactly its two axes, ranging from 0 to 1.

Finally, the AP is the integration of the PRC curve over the interval from $Recall = 0$ to $Recall = 1$. Just as the name of the AP replies, it is a more robust measurement of the model performance.

2) Object Scale: The *object scale* is a measurement of ground truth bounding box $\boldsymbol{Bx}^{g\prime}$ size and its computational detail is illustrated in the following.

Since the object detection framework preprocesses the input image by resizing the shorter edge to a fixed length (600 pixels) and resizing the other edge with the same scale, we first resize $\boldsymbol{Bx}^{g\prime}$ to get $\boldsymbol{Bx}^{g}$

$$\boldsymbol{Bx}^{g} = (x, y, w, h). \tag{25}$$

Then, we get *object scale* with the formula

$$\text{Scale} = \sqrt{w \times h}. \tag{26}$$

Obviously, this is a relative measurement of object size. In addition, for a data set, the mean value and the standard deviation of the scales of each category objects are calculated.

3) Improvement–Scale Diagram: This graph takes the *object scale* mean value, namely *mean resized scale*, as its horizontal axis and the AP improvement (compared with the backbone network) as its vertical axis. Each point in the *improvement–scale diagram* represents a category. After all category points are ready, a trend line is drawn to pinpoint their statistical trend.

This graph is designed to clearly exhibit the relationship between the AP improvement and the object scale to evaluate specific scale detection ability.

4) Intersection Over Union: To evaluate our method, detection results are reduced by removing redundant boxes using IoU between the predicted box $\boldsymbol{Bx}^{p}$ and the ground truth box $\boldsymbol{Bx}^{g}$. The IoU is computed by

$$\text{IoU} = \frac{\boldsymbol{Bx}^{p} \cap \boldsymbol{Bx}^{g}}{\boldsymbol{Bx}^{p} \cup \boldsymbol{Bx}^{g}}. \tag{27}$$

D. Experimental Verification on HRRSD

After training on HRRSD, the GACL Net is evaluated on the test set. Some detection results are shown in Fig. 9, with the first column representing backbone (CaffeNet) detections and the right column representing GACL Net detections. The localization improvement is conspicuous after utilizing the GACL Net. This phenomenon is apparent on the airplanes in this figure.

Different category APs between our method and the backbone network are compared in Table V. In order to further verify the validity of our proposed model, we compare the GACL Net and the backbone network under different CNN conditions.

The first CNN type is the CaffeNet [9] whose detection results are shown in the "CaffeNet" column of Table V. The other two types are the VGG16 [10] and its simplified version. Their detection results are shown in the "VGG16" column and the "VGG_CNN_M_1024" column of this table, respectively. Moreover, the columns with "GACL" prefix (such as "GACL CaffeNet") mean the detection results of corresponding GACL localization models. In fact, these three backbone CNNs are ranked from simple to complex as: CaffeNet, VGG_CNN_M_1024, and VGG16.

As a visualization form of Table V, the relationships between AP improvements by GACL Nets and resized scales of different categories are shown through the *improvement–scale diagram* in Fig. 10. The horizontal axis is the AP improvement of GACL Net compared with the corresponding

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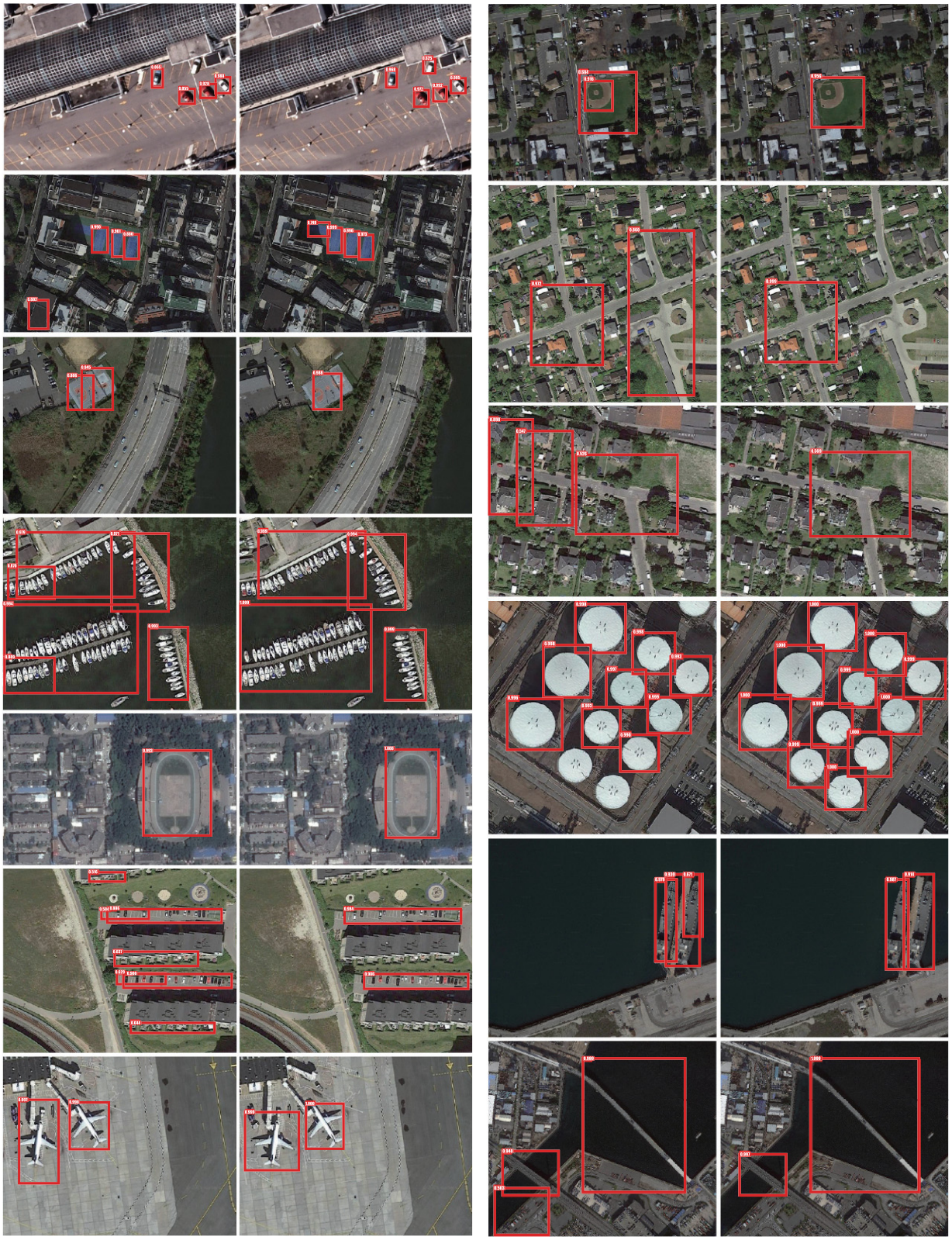


Fig. 9. Detection results comparison between the backbone network and the proposed GACL Net. Numbers on those boxes are their recognition scores.

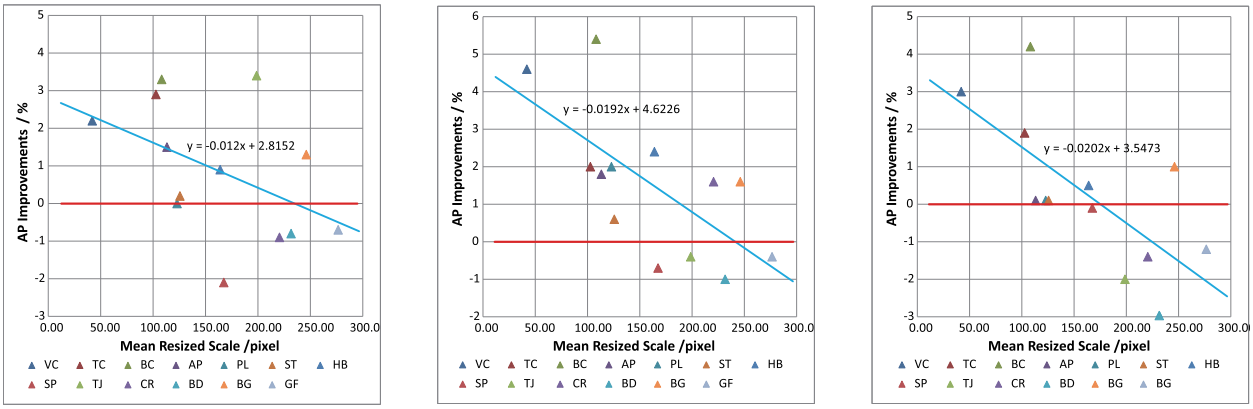
backbone network AP which is represented by the red line. The legends at the bottom of the subfigures are the abbreviations of different categories which are represented by different color triangles.

It is clearly shown in these subfigures that our model works better in detecting object categories with smaller scales. In this case, the effectiveness of our method for object categories with small scales is verified.

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TABLE V
EXPERIMENTAL VERIFICATION OF THE PROPOSED LOCALIZATION MODEL ON HRRSD

	CaffeNet /%	GACL CaffeNet /%	Changed GACL CaffeNet /%	VGG_CNN_M_1024 /%	GACL VGG_CNN_M_1024 /%	VGG16 /%	GACL VGG16 /%
airplane	80.3	81.8	81.7	83.3	85.1	88.8	88.9
baseball diamond	82.9	82.1	81.3	83.6	82.6	84.3	81.3
basketball court	31.6	34.9	35.0	36.7	42.1	44.0	48.2
bridge	73.5	74.8	75.3	75.1	76.7	76.3	77.3
crossroad	65.8	64.9	66.3	67.1	68.7	69.3	67.9
ground track field	89.8	89.1	88.9	90.0	89.6	90.1	88.9
harbor	77.8	78.7	78.2	76.0	78.4	79.7	80.2
parking lot	34.1	34.1	33.8	37.5	39.5	38.3	38.4
ship	76.1	74.0	73.2	75.0	74.3	78.2	78.1
storage tank	79.6	79.8	80.0	79.8	80.4	80.1	80.2
T junction	33.9	37.3	37.0	39.2	38.8	42.1	40.1
tennis court	70.1	73.0	72.7	75.0	77.0	77.5	79.4
vehicle	39.9	42.1	43.1	46.1	50.7	50.7	53.7
mAP	64.3	**65.1**	**65.1**	66.5	**68.0**	69.2	**69.4**



(a) GACL CaffeNet (b) GACL VGG CNN M 1024 (c) GACL VGG16

Fig. 10. Visualization of the relationship between GACL Net detection results and object scales on HRRSD, as using different CNNs for GACL Net. (a)–(c) Experimental results using the backbone CNNs of CaffeNet, VGG CNN M 1024, and VGG16, respectively. The red line is the backbone CNN and the blue line represents the GACL Net with the same CNN as the backbone. Different colors of triangles represent different object categories: {airplane: AP; baseball diamond: BD; basketball court: BC; bridge: BG; crossroad: CR; ground track field: GF; harbor: HB; parking lot: PL; ship: SP; storage tank: ST; T junction: TJ; tennis court: TC; vehicle: VC}.

TABLE VI
EXPERIMENTAL VERIFICATION OF THE PROPOSED LOCALIZATION MODEL ON NWPU VHR-10

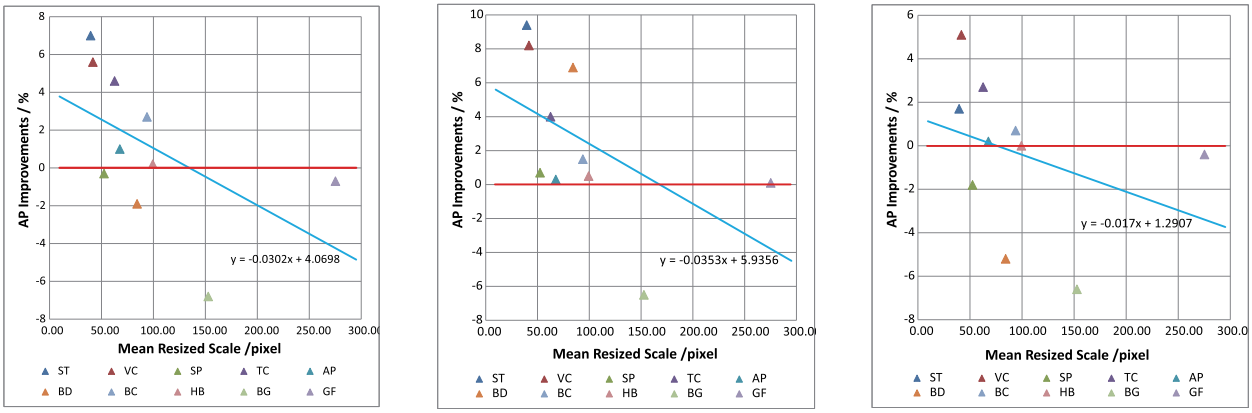
	CaffeNet /%	GACL CaffeNet /%	VGG_CNN_M_1024 /%	GACL VGG_CNN_M_1024 /%	VGG16 /%	GACL VGG16 /%
airplane	88.6	89.6	90.0	90.3	90.2	90.4
baseball diamond	90.2	88.3	90.0	96.9	96.4	91.2
basketball court	79.2	81.9	80.7	82.2	81.1	81.8
bridge	79.7	72.9	79.2	72.7	79.4	72.8
ground track field	99.6	98.9	99.3	99.4	99.3	98.9
harbor	90.7	90.9	90.4	90.9	90.8	90.8
ship	63.8	63.5	62.8	63.5	63.9	62.1
storage tank	46.4	53.5	44.5	53.9	51.9	53.6
tennis court	75.4	80.0	76.9	80.9	78.3	81.0
vehicle	49.4	55.3	49.7	57.9	51.1	56.2
mAP	76.3	**77.5**	76.4	**78.9**	**78.2**	77.9

E. Experimental Verification on NWPU VHR-10

Different category APs on the publicly available data set NWPU VHR-10 between our methods and the backbone networks are compared in Table VI. In this table, different backbone CNNs and corresponding GACL Nets are validated.

The visualizations of these detections are shown in Fig. 11. It is obvious that the relationship between the AP improvement of GACL Net and the resized object scale remains the same with different CNNs on NWPU VHR-10: the proposed localization model works well for object categories with small

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(a) GACL CaffeNet (b) GACL VGG_CNN_M_1024 (c) GACL VGG16

Fig. 11. Visualization of the relationship between GACL Net detection results and object scales on NWPU VHR-10, as using different CNNs for GACL Net. (a)–(c) Experimental results using the backbone CNNs of CaffeNet, VGG CNN M 1024, and VGG16, respectively. The red line is the backbone CNN and the blue line represents the GACL Net with the same CNN as the backbone. Different colors of triangles represent different object categories: {airplane: AP; baseball diamond: BD; basketball court: BC; bridge: BG; crossroad: CR; ground track field: GF; harbor: HB; parking lot: PL; ship: SP; storage tank: ST; T junction: TJ; tennis court: TC; vehicle: VC}.

TABLE VII
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON HRRSD

	BOW [44] /%	SSCBoW [23] /%	FDDL [18] /%	COPD [16] /%	Transformed CNN [45] /%	RICNN [4] /%	YOLO [46] /%	Fast RCNN [28] /%	GACL Fast RCNN /%	Faster RCNN [25] /%	GACL Faster RCNN /%
Airplane	36.80	59.29	30.49	62.53	77.46	78.05	84.6	83.3	85.1	90.8	**90.8**
Baseball Diamond	7.01	34.96	20.35	52.18	57.60	59.64	62.2	83.6	82.6	86.9	**87.2**
Basketball Court	4.36	7.94	2.53	9.56	17.97	22.95	41.3	36.7	42.1	47.9	**49.7**
Bridge	7.28	9.30	10.08	7.12	19.98	27.44	79.0	75.1	76.7	85.5	**85.6**
Crossroad	12.17	11.59	18.22	8.57	25.90	26.63	43.4	67.1	68.7	**88.6**	88.2
Ground Track Field	15.69	7.56	14.98	72.40	76.28	78.00	**94.4**	90.0	89.6	90.6	90.7
Harbor	42.12	53.50	19.27	47.80	54.13	47.81	74.4	76.0	78.4	89.4	**89.7**
Parking Lot	5.29	6.84	7.41	17.44	16.57	20.45	45.8	37.5	39.5	63.3	**65.3**
Ship	39.36	37.55	33.16	45.26	49.68	56.54	78.5	75.0	74.3	88.5	**88.5**
Storage Tank	51.48	26.30	72.72	50.06	79.10	80.95	72.4	79.8	80.4	88.7	**89.2**
T Junction	0.72	0.81	1.86	1.55	2.42	9.33	46.8	39.2	38.8	**75.1**	75.0
Tennis Court	12.53	12.16	15.99	58.09	70.83	66.37	67.6	75.0	77.0	80.7	**80.8**
Vehicle	11.39	30.46	4.02	32.70	41.27	51.95	65.1	46.1	50.7	84.0	**86.9**
Mean AP	18.94	22.94	19.31	35.79	45.32	48.16	65.8	66.5	68.0	81.5	**82.1**

scales. These experiments also verify the effectiveness of our method for object categories with small scales.

F. Comparison With Other Methods

To prove the validity of our method, it is compared with nine existing methods, including four traditional state-of-the-art methods and five CNN-based methods. These four methods are the BoW [44], the SSCBoW [23], the FDDL [18], and the COPD [16]. Those five CNN-based methods are the transformed CNN models from AlexNet [45], the RICNN [4], the YOLO [46], the Fast RCNN [28], and the Faster RCNN [25].

The contrast experiments on HRRSD and NWPU VHR-10 are, respectively, shown in Tables VII and VIII. The detection results in these tables show that our model gets the best mAP on both the data sets. Moreover, based on the Fast RCNN that takes SS for region proposal, GACL Net improves mAPs by 1.5% and 2.5% on HRRSD and NWPU VHR-10, respectively. Based on the Faster RCNN that takes RPN for the region proposal, the GACL Net improves mAPs by 0.6% and 0.4% on HRRSD and NWPU VHR-10, respectively.

G. GACL Net Variation

In addition, we wonder whether there would be a greater AP improvement if a change is made in the proposed model. As shown in Fig. 8, in the global-gating process, the same feature vector is used as the gates of the x-direction and the y-direction features. We try using two different vectors as the gates. We have explored this variation of GACL Net and got the experimental results as shown in column "Changed GACL CaffeNet" in Table V.

The AP improvement is visualized in Fig. 12. It is obvious that the APs after such a change are similar to the original. Moreover, this change takes one more global feature vector and thus requires another fully connected layer, which takes additional computational cost.

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TABLE VIII
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON NWPU VHR-10

	BOW [44] /%	SSCBoW [23] /%	FDDL [18] /%	COPD [16] /%	Trans-formed CNN [45] /%	RICNN [4] /%	YOLO [46] /%	Fast RCNN [28] /%	GACL Fast RCNN /%	Faster RCNN [25] /%	GACL Faster RCNN /%
Airplane	24.96	50.61	29.15	62.25	70.14	88.35	83.0	90.0	90.3	90.7	**90.9**
Baseball Diamond	9.03	43.49	25.76	83.27	83.61	88.12	84.3	90.0	**96.9**	89.2	88.8
Basketball Court	3.22	14.96	3.58	36.25	46.80	58.45	53.5	80.7	82.2	87.5	**88.4**
Bridge	12.16	12.49	21.54	14.79	45.38	61.51	85.0	79.2	72.7	80.9	**86.1**
Ground Track Field	7.77	10.07	20.10	85.31	81.20	86.73	62.8	99.3	99.4	**100.0**	99.8
Harbor	52.98	58.33	25.39	55.27	62.28	68.60	78.7	90.4	**90.9**	89.8	90.0
Ship	58.49	50.84	37.64	68.87	63.70	77.34	84.4	62.8	63.5	**89.9**	89.8
Storage Tank	63.18	33.37	77.00	63.71	84.33	**85.27**	81.9	44.5	53.9	67.3	68.5
Tennis Court	4.72	0.33	2.75	32.08	35.46	40.83	**85.0**	76.9	80.9	78.6	79.7
Vehicle	9.14	33.61	4.47	44.03	44.80	71.10	70.0	49.7	57.9	88.0	**88.5**
Mean AP	24.57	30.81	24.74	54.58	61.77	72.63	76.8	76.4	78.9	86.2	**86.6**

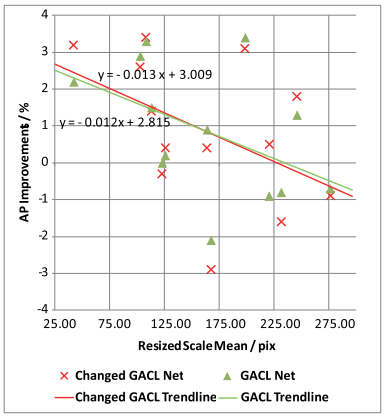


Fig. 12. Visualization of the trial experiments over different GFG strategies through the *improvement–scale diagram*. The horizontal axis is the AP improvement of GACL Net compared with the backbone network AP. The red line represents the changed GACL Net and the green one is the GACL Net.

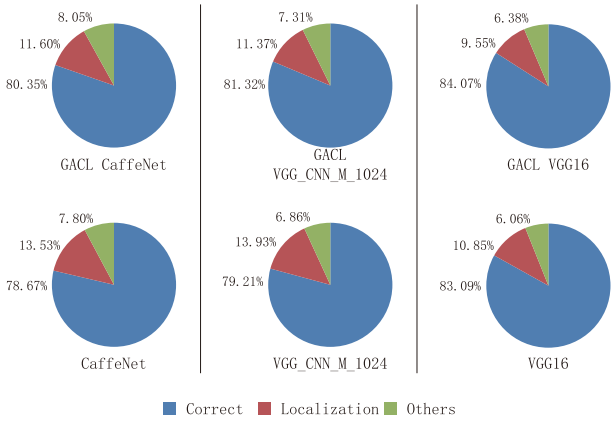


Fig. 13. Visualization of localization improvement. Each pie chart shows the percentage of localization errors in ground truth boxes.

H. Localization Improvement Verification

To clearly display the localization improvement brought by the GACL Net, we analyze the percentages of localization errors in the ground truth boxes of HRRSD. The results are shown in Fig. 13. Different colors represent different detection result types, which are illustrated as follows.

1) *Correct:* Correctly detected boxes with proper category and IoU greater than 0.5.
2) *Localization:* Missed boxes with proper category but IoU between 0.1 and 0.5.
3) *Others:* Missed boxes with error category or IoU lower than 0.1.

It can be seen in this figure that, after using the GACL Net, the localization error rate is reduced by 1.93% on CaffeNet, 2.56% on VGG_CNN_M_1024, and 1.30% on VGG16. In addition, the GACL Net also improves the correct rate by 1.68% on CaffeNet, 2.11% on VGG_CNN_M_1024, and 0.98% on VGG16. This demonstrates that the GACL Net can significantly reduce localization errors on different CNNs.

I. Ablation Experiments

To illustrate the function of each part in the GACL Net, we launch the ablation experiments in Table IX. In these

TABLE IX
ABLATION EXPERIMENTS ON HRRSD

	ACP	GFG	mAP /%	vehicle /%	tennis court /%	basketball court /%
1	✓	✓	**65.1**	**42.1**	**73.0**	**34.9**
2	✓		64.8	41.8	71.8	34.5
3			64.3	39.9	70.1	31.6

experiments, we evaluate the model on the data set HRRSD with the backbone of CaffeNet. Compared with experiment 3, experiment 2 with ACP improves the mAP by 0.5% and experiment 1 with ACP and GFG improves the mAP by 0.8%. Moreover, since the object categories with small scales are the most concerned, we evaluate with the three smallest object categories (vehicle, tennis court, and basketball court) and find that the GACL Net improves the APs of these categories by 2.2%∼3.3%.

VI. CONCLUSION

This article proposes a novel localization model, the GACL Net, to improve the performance of small-scale object detections in remote sensing images. In this model, the GFG uses the global features to guide the channel attention of the local convolutional features, and the ACP takes the single-axis

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pooling process to avoid coordinate prediction disturbance. We observe the relationship between the AP improvements by the GACL Net and the *object scale* of different categories and find that object categories with small scales always get more AP improvements with the GACL Net. Furthermore, we implement our model on different CNN baselines and different data sets to verify that our model can work well for object categories with small scales.

REFERENCES

[1] S. Valero, J. Chanussot, J.-A. Benediktsson, H. Talbot, and B. Waske, "Advanced directional mathematical morphology for the detection of the road network in very high resolution remote sensing images," *Pattern Recognit. Lett.*, vol. 31, no. 10, pp. 1120–1127, Jul. 2010.

[2] F. Chen, R. Ren, T. Van de Voorde, W. Xu, G. Zhou, and Y. Zhou, "Fast automatic airport detection in remote sensing images using convolutional neural networks," *Remote Sens.*, vol. 10, no. 3, p. 443, Mar. 2018.

[3] H. Grabner, T. T. Nguyen, B. Gruber, and H. Bischof, "On-line boosting-based car detection from aerial images," *J. Photogramm. Remote Sens.*, vol. 63, no. 3, pp. 382–396, May 2008.

[4] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[5] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

[6] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[7] L. Huang, Y. Yang, Y. Deng, and Y. Yu, "Densebox: Unifying landmark localization with end to end object detection," Sep. 2015, *arXiv:1509.04874*. [Online]. Available: https://arxiv.org/abs/1509.04874

[8] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," Sep. 2017, *arXiv:1709.01507*. [Online]. Available: https://arxiv.org/abs/1709.01507

[9] Y. Jia *et al.*, "Caffe: Convolutional architecture for fast feature embedding," in *Proc. 22nd ACM Int. Conf. Multimedia*, Nov. 2014, pp. 675–678.

[10] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," Sep. 2014, *arXiv:1409.1556*. [Online]. Available: https://arxiv.org/abs/1409.1556

[11] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[12] D. G. Lowe, "Object recognition from local scale-invariant features," in *Proc. IEEE Int. Conf. Comput. Vis.*, vol. 2, Sep. 1999, pp. 1150–1157.

[13] Q. Li, G. Wang, J. Liu, and S. Chen, "Robust scale-invariant feature matching for remote sensing image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 287–291, Apr. 2009.

[14] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[15] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, vol. 1, no. 1, pp. 886–893.

[16] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[17] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.

[18] J. Han *et al.*, "Efficient, simultaneous detection of multi-class geospatial targets based on visual saliency modeling and discriminative learning of sparse coding," *ISPRS J. Photogram. Remote Sens.*, vol. 89, pp. 37–48, Mar. 2014.

[19] W. Zhang, X. Sun, K. Fu, C. Wang, and H. Wang, "Object detection in high-resolution remote sensing images using rotation invariant parts based model," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 74–78, Jan. 2014.

[20] Y. Li, X. Sun, H. Wang, H. Sun, and X. Li, "Automatic target detection in high-resolution remote sensing images using a contour-based spatial model," *IEEE Trans. Geosci. Remote Sens.*, vol. 9, no. 5, pp. 886–890, Sep. 2012.

[21] L. Fei-Fei and P. Perona, "A Bayesian hierarchical model for learning natural scene categories," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 2. Jun. 2005, pp. 524–531.

[22] D. Zhang, J. Han, G. Cheng, Z. Liu, S. Bu, and L. Guo, "Weakly supervised learning for target detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 701–705, Apr. 2015.

[23] H. Sun, X. Sun, H. Wang, Y. Li, and X. Li, "Automatic target detection in high-resolution remote sensing images using spatial sparse coding bag-of-words model," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 1, pp. 109–113, Jan. 2012.

[24] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006.

[25] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jul. 2017.

[26] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. Lecun, "Overfeat: Integrated recognition, localization and detection using convolutional networks," Dec. 2013, *arXiv:1312.6229*. [Online]. Available: https://arxiv.org/abs/1312.6229

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[28] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 1440–1448.

[29] X. Wang, A. Shrivastava, and A. Gupta, "A-fast-RCNN: Hard positive generation via adversary for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2606–2615.

[30] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.

[31] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2016, pp. 21–37.

[32] C. Y. Fu, W. Liu, A. Ranga, A. Tyagi, and A. C. Berg, "DSSD : Deconvolutional single shot detector," Jan. 2017, *arXiv:1701.06659*. [Online]. Available: https://arxiv.org/abs/1701.06659

[33] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, to be published.

[34] H. Zhu, X. Chen, W. Dai, K. Fu, Q. Ye, and J. Jiao, "Orientation robust object detection in aerial images using deep convolutional neural network," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 3735–3739.

[35] F. Hu, G.-S. Xia, J. Hu, and L. Zhang, "Transferring deep convolutional neural networks for the scene classification of high-resolution remote sensing imagery," *Remote Sens.*, vol. 7, no. 11, pp. 14680–14707, Nov. 2015.

[36] S. Gidaris and N. Komodakis, "Object detection via a multi-region and semantic segmentation-aware CNN Model," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1134–1142.

[37] S. Gidaris and N. Komodakis, "Locnet: Improving localization accuracy for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 789–798.

[38] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Sep. 2013.

[39] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[40] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3431–3440.

[41] S. Reed, K. Sohn, Y. Zhang, and H. Lee, "Learning to disentangle factors of variation with manifold interaction," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 1431–1439.

[42] M. Cramer, "The DGPF-test on digital airborne camera evaluation–overview and test design," *Photogrammetrie-Fernerkundung-Geoinformation*, vol. 2010, no. 2, pp. 73–82, May 2010.

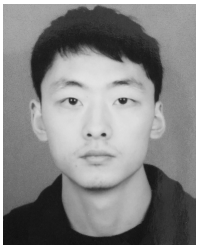
[43] E. Ohn-Bar and M. M. Trivedi, "Multi-scale volumes for deep object detection and localization," *Pattern Recognit.*, vol. 61, pp. 557–572, Jan. 2016.

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[44] S. Xu, T. Fang, D. Li, and S. Wang, "Object classification of aerial images with bag-of-visual words," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 2, pp. 366–370, Apr. 2010.
[45] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[46] J. Redmon and A. Farhadi, "Yolo9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 7263–7271.



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